

the complex part of managing resistance-capacitance.

Again, there are local interconnects at the device level, accomplished by middle-of-line process. Global interconnects are done by back-end-of-line and are prone to resistance-capacitance delays. Today, at lower nodes, back-end-of-line employs multiple patterning, which requires extra deposition and etching with every pattern, thus increasing the cost of production.

Technically, multiple patterning can still be viable at 7nm. However, the industry is looking at extreme ultraviolet (EUV) lithography to reduce that cost. With EUV, back-end-of-line process can be done with single exposure and throughput can be as good as ~150 wafers per hour. But for EUV lithography, foundries are dependent on semiconductor equipment manufacturing companies. It has developed a technology to fabricate test patterns with minimum line width of 1.5nm, which can be used to test metrological equipment with ultra-high precisions. Yield and patterning technologies are important.

Yield

Standard architecture of high-performance logic chips these days assumes that all transistors work, and we are talking about chips that would contain several billion transistors each. In terms of manufacturing and reliability, integrated semiconductor devices are amazing and put mechanical devices to shame—billions of components connected in a complicated way, and barring disaster, these things can all work for many years at a time without failure.

To be manufacturable, chip makers need to have the yield of good chips high enough to cover the manufacturing costs at a reasonable

price point. That typically means yield rates over at least 30 per cent.

Patterning technology

All chips in recent years and even their prototype 10nm node products are patterned using photolithography, based on a light source with a wavelength of 193nm. Manufacturers have relied on pattern features down to one-twentieth of the free-space wavelength of light, including immersion lithography, optical phase control, exotic photochemistry and multiple patterning.

Finally, it is time to switch to a new light source of 13.5nm wavelength, which is the so-called EUV. This has been in the planning stages for years, with prototype EUV 300mm wafer systems. However, changing to new wavelengths and therefore new processing chemistry and procedures is fraught with challenges.

EUV lithography

EUV lithography represents a major technological advancement for semiconductor manufacturing. Current lithographic techniques utilise deep ultraviolet (DUV) light sources that produce wavelengths of 248nm or 193nm. EUV light used in EUV lithography has a wavelength of 13.5nm, a full order of magnitude shorter. This is important because

imaging capabilities or resolutions of a lithography system are proportional to and limited by the wavelength of light used.

EUV remains a consistent challenge, and though manufacturers swear they can solve the power problem, the difficulties are staggering. Gate-all-around is an attractive method for improving transistor performance, but manufacturers are still struggling to hit reliability and dimension challenges.

Materials like graphene or carbon nanotubes meanwhile would not debut for a decade or more in complex CMOS logic devices.

EUV light source produces short wavelength light in a system called laser produced plasma. Such light sources utilise a high-power laser to create high-energy plasma that emits short wavelength light inside a vacuum chamber. There are multiple ways that EUV differs, mostly associated with methods to create and transport the short wavelength light.

Traditional lenses cannot be used with EUV as these absorb light. A mirror with a highly specialised coating, called multilayer mirror, must be used. Even these special mirrors absorb about 30 per cent of the light, so it is advantageous to use as few as possible. Any gas in the light path such as air or nitrogen will also absorb the light, thus the entire light path is inside a vacuum chamber. Primary EUV tool maker, ASML, projects EUV at 5nm node would require a higher numerical aperture than currently available and multiple patterning to a greater degree than immersion lithography at 20nm node.

Immersion lithography is still more than four times faster than EUV (275 WPH versus 65 WPH as detailed below), due to source power limitations. Hence, multiple

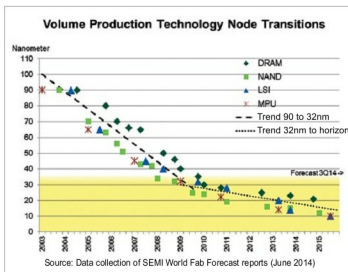


Fig. 3: Nanometer versus years